

Constructing (Co)inductive Types using an Impredicative Universe, or Parametric Sizes.

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(Co)inductive Types

Proof assistants (Agda, Rocq, Lean, ...) allow you to define mathematical objects as **inductive** or **coinductive** types.

Example. The **inductive** type $\mathbb{N}$ is **freely** generated by **constructors**:

$$0 : \mathbb{N},$$

$$\text{suc} : \mathbb{N} \rightarrow \mathbb{N}.$$

The **coinductive** type $\text{Stream } A$ is **cofreely** generated by **destructors**:

$$\text{head} : \text{Stream } A \rightarrow A,$$

$$\text{tail} : \text{Stream } A \rightarrow \text{Stream } A.$$

Inductive	initial	eager data
Coinductive	terminal	lazy computation

Usually these are **primitive** built-ins; can we **define** them instead?

Why Construct (Co)inductive Types?

We want to **bridge** two worlds:

- a **minimalistic** theory: easy to **model externally**;
- a **feature-rich** theory: easy to **prove internally**.

Constructing **(co)inductive** types in a simple core combines the **convenience** of the **rich** theory with the **trust** of the **simple** theory.

Two formulations are **used in practice**, but currently **flawed**:

- **impredicative encodings**: used in **Calculus of Constructions** and **System F**, but the universal property is **weak**;
- **sized types**: implemented in **Agda**, but **inconsistent**.

We give refinements of both formulations.

Both refinements use **function extensionality**, yet stay compatible with both **UIP** *and* **univalence**.

UIP / Univalence

Two **incompatible** ways to treat **identity** of types:

- **UIP**: any type is a set (**0-truncated**): all $\alpha, \alpha' : a =_A a'$ are equal;
- **univalence** (HoTT): the canonical map $(A =_{\mathcal{U}} B) \rightarrow (A \simeq B)$ is an equivalence ($\mathcal{U}$ is **not 0-truncated**).

We want to be **compatible** with both, so we accept **neither**.

Price: without **UIP** we can have **infinitely** many levels

$$X, X' : \mathcal{U}, \quad f, f' : X \rightarrow X', \quad \phi, \phi' : f = f', \quad \Phi, \Phi' : \phi = \phi', \quad \dots$$

So types are $(\infty, 1)$ -categories, with **infinitely** much coherence data.

Impredicative Universe

(Im)predicative

A definition is:

- **predicative** if we **only** quantify over objects that **exist before**;
- **impredicative** if we **might** quantify over objects that **include itself**;
for example: the **greatest** fixed point is the join of **all** fixed points.

For a universe $\mathcal{U}$ of **small** types: is $\Pi_{X:\mathcal{U}} X$ also **small**?

- **predicative**: quantifying over all **small** types produces a **large** type;
(Martin-Löf type theory, Agda);
- **impredicative**: quantifying over all **small** types produces a **small** type.
(Calculus of Constructions, System F, Prop in Rocq and Lean).

Reassuringly, an **impredicative universe** is **consistent** with both **UIP** and **univalence** (realizability: **(cubical) assembly models**).

The Classic Impredicative Encoding

In **second-order logic**, all connectives can be defined using only $\rightarrow$ and $\forall$:

$$\begin{aligned}\perp &:= \forall_{X:\text{Prop}} X, \\ \top &:= \forall_{X:\text{Prop}} (X \rightarrow X), \\ A \vee B &:= \forall_{X:\text{Prop}} ((A \rightarrow X) \rightarrow (B \rightarrow X) \rightarrow X), \\ A \wedge B &:= \forall_{X:\text{Prop}} ((A \rightarrow B \rightarrow X) \rightarrow X), \\ \exists_{x:A} B[x] &:= \forall_{X:\text{Prop}} (\forall_{x:A} (B[x] \rightarrow X) \rightarrow X), \\ (a =_A b) &:= \forall_{X:A \rightarrow \text{Prop}} (X a \rightarrow X b).\end{aligned}$$

Under **Curry–Howard**, this gives definitions for $0, 1, +, \times, \Sigma, =$ using only Π and $\mathcal{U}$, and more generally for any **inductive type**, for example:

$$\mathbb{N} := \Pi_{X:\mathcal{U}} (X \rightarrow (X \rightarrow X) \rightarrow X).$$

The closed terms are correct, but these encodings are only **weakly** initial: we have **iteration** but **not induction** (dependent elimination).

Impredicative Encodings: A Brief History

Steadily stronger **universal properties**, needing steadily more **type formers**:

1940s	Church	weak	inductive	$\Pi, \mathcal{U}$
1990s	Girard–Taylor–Lafont / Geuvers	weak	(co)inductive	$\Pi, \mathcal{U}$
2011	Shulman (blog)	weak	higher inductive	$=, \Pi, \mathcal{U}$
2018	Awodey–Frey–Speight	n -truncated	higher inductive	$=, \Sigma, \Pi, \mathcal{U}$
2023	Shulman (blog) / Ripoll Echeveste [†]		inductive	$\mathbb{N}, =, \Sigma, \Pi, \mathcal{U}$
2025	Volpe [†]		higher inductive	$\mathbb{N}, =, \Sigma, \Pi, \mathcal{U}$
2025	Bronsveld–Geuvers–Van der Weide [†]	0-truncated	(co)inductive	$=, \Sigma, \Pi, \mathcal{U}$
this work			(co)inductive	$\mathbb{N}, =, \Sigma, \Pi, \mathcal{U}$

However although we need $\mathbb{N} : \text{Type}$, we do **not** need $\mathbb{N} : \mathcal{U}$.

[†] MSc theses we supervised in **Amsterdam** and **Nijmegen**.

Overview

Plan: phrase everything categorically; then

inductive $\Leftrightarrow$ initial algebra, coinductive $\Leftrightarrow$ terminal coalgebra

are formally dual.

Most results will dualize for free and we isolate what few things do *not*.

We start with the 0-truncated / UIP case of 1-categories before moving to the general case of $(\infty, 1)$ -categories.

Initial Algebras and Terminal Coalgebras

Categorically, for a functor $F : \text{Type} \rightarrow \text{Type}$:

- $(\mu F, \text{cons})$ is **initial** among **algebras** $(X : \mathcal{U}, c : F X \rightarrow X)$,
- $(\nu F, \text{des})$ is **terminal** among **coalgebras** $(X : \mathcal{U}, d : X \rightarrow F X)$.

Example. The inductive type $(\mathbb{N}, (0, \text{suc}))$ is **initial** among pairs:

$$(X : \mathcal{U}, (z, s) : 1 + X \rightarrow X).$$

In detail: for any $(X, (z, s))$, there exists a map $\text{iter} : \mathbb{N} \rightarrow X$ with:

$$\begin{array}{ccc} 1 + \mathbb{N} & \xrightarrow{1 + \text{iter}} & 1 + X \\ (0, \text{suc}) \downarrow & \text{iter}_\beta & \downarrow (z, s) \\ \mathbb{N} & \xrightarrow{\text{iter}} & X \end{array} \quad (\text{iteration})$$

Moreover, $(\text{iter}, \text{iter}_\beta) : (\mathbb{N}, (0, \text{suc})) \rightarrow (X, (z, s))$ is **unique**. (induction)

$(\text{Stream } A, (\text{head}, \text{tail}))$ is **terminal** among $(X : \mathcal{U}, (h, t) : X \rightarrow A \times X)$.

With UIP: Carving Out the Initial Algebra

The classical encoding quantifies over **all** algebras and returns the carrier:

$$\mu_0 F := \Pi_{X:\mathcal{U}, c:F X \rightarrow X} X.$$

By **impredicativity** $\mu_0 F : \mathcal{U}$, and it is a **weakly initial** algebra:

$$\begin{aligned} \text{cons}_0 : F(\mu_0 F) &\rightarrow \mu_0 F, & \text{iter}_0 : \Pi_{X:\mathcal{U}, c:F X \rightarrow X} (\mu_0 F \rightarrow X), \\ \text{cons}_0 u' X c &:= c(F(\text{iter}_0 X c) u'), & \text{iter}_0 X c u &:= u X c. \end{aligned}$$

Not initial (too many morphisms out). **Fix:** keep only those $u : \mu_0 F$ where iter_0 acts **natural**:

$$\begin{array}{ccc} & \text{iter}_0 X c & \nearrow \\ (\mu_0 F, \text{cons}_0) & & (X, c) \\ & \searrow \text{iter}_0 X' c' & \downarrow f \\ & & (X', c') \end{array}$$

Then $\mu_1 F := \Sigma_{u:\mu_0 F} (f(\text{iter}_0 X c u) = \text{iter}_0 X' c' u)$ is **initial**.

Smallness Forbids Large Elimination

$\mu_0 F := \Pi_{X:\mathcal{U}, c:F X \rightarrow X} X$ and $\mu_1 F := \Sigma_{u:\mu_0 F} (u \text{ natural})$ are always small, even when $F X$ is large.

Hence large elimination (into large algebras) would be inconsistent:

Example. (the truncation/reflection $\|A\|_{\mathcal{U}} : \mathcal{U}$) The universal small type $\|A\|_{\mathcal{U}}$ for a large type A is freely generated by a constructor:

$$|-| : A \rightarrow \|A\|_{\mathcal{U}}.$$

Large elimination would give a map $\|A\|_{\mathcal{U}} \rightarrow A$, hence $A \simeq \|A\|_{\mathcal{U}}$.

So, every type would be essentially small: even $\mathcal{U}$, which is inconsistent.

So impredicative encodings have only small elimination.

With UIP: Collapsing Out the Terminal Coalgebra

Dually, quantify over all $F : \mathcal{U} \rightarrow \mathcal{U}$ coalgebras and return a state:

$$\nu_0 F := \left\| \Sigma_{X:\mathcal{U}, d:X \rightarrow F X} X \right\|_{\mathcal{U}}. \quad (\Sigma_{X,d} X \text{ is large, so reflect it into } \mathcal{U})$$

By truncation $\nu_0 F : \mathcal{U}$, and it is a weakly terminal coalgebra:

$$\begin{aligned} \text{des}_0 : \nu_0 F &\rightarrow F(\nu_0 F), & \text{coiter}_0 : \Pi_{X:\mathcal{U}, d:X \rightarrow F X} (X \rightarrow \nu_0 F), \\ \text{des}_0 |X, d, x| &:= F(\text{coiter}_0 X d)(dx), & \text{coiter}_0 X d x &:= |X, d, x|. \end{aligned}$$

Not terminal (too many morphisms in). Fix: identify states joined by a coalgebra map f , making coiter_0 natural:

$$\begin{array}{ccc} (X, d) & \xrightarrow{\text{coiter}_0 X d} & (\nu_0 F, \text{des}_0) \\ f \downarrow & & \uparrow \\ (X', d') & \xrightarrow{\text{coiter}_0 X' d'} & \end{array}$$

Then $\nu_1 F := \nu_0 F / (\text{coiter}_0 X d x = \text{coiter}_0 X' d' (f x))$ is terminal.

Without UIP: Wild Categories

Drop UIP: **naturality** is now **data**, *not* a **proposition**.

More worryingly, $\mu_1 F$ and $\nu_1 F$ are no longer **initial** and **terminal**.

Types form an $(\infty, 1)$ -category, which is too big to write down, so we keep only **finitely** much of the data:

Definition. A **wild category** has a type $\mathcal{C}$ of **objects**, for every $X, X' : \mathcal{C}$ a type $X \rightarrow_{\mathcal{C}} X'$ of **morphisms** (*not a set*), **identities**, **composition**, and equalities for **unitality** and **associativity**, and **nothing higher**.

Likewise **wild functors** and **wild natural transformations** carry their **preservation** and **naturality** data as equalities, and **nothing higher**.

$\mathcal{C}$ is **strict** if **unitality** and **associativity** hold **definitionally**.

Example. $\mathcal{U}$ (small types) and **Type** (all types), with functions and equalities, are **strict wild categories** (using η).

Do Algebras even Form a Wild Category?

We can define **identity** and **composition** morphisms, using that F preserves **id** and **composition**:

$$\begin{array}{ccc}
 F X & \xrightarrow{\quad F \text{id}_X \quad} & F X \\
 \downarrow c & \searrow \scriptstyle (\text{pres}_{\text{id}_X}^F)^{-1} & \downarrow c \\
 & \text{id}_{F X} & \\
 & \text{refl}_c & \\
 X & \xrightarrow{\quad \text{id}_X \quad} & X
 \end{array}$$

(identity)

$$\begin{array}{ccccc}
 F X & \xrightarrow{\quad F(f \cdot f') \quad} & & F X'' & \\
 \downarrow c & \searrow \scriptstyle F f & & \nearrow \scriptstyle F f' & \downarrow c'' \\
 & & F X' & & \\
 & \searrow \scriptstyle \phi & & \nearrow \scriptstyle \phi' & \\
 X & \xrightarrow{\quad f \quad} & X' & \xrightarrow{\quad f' \quad} & X''
 \end{array}$$

(composition)

Warning: unitality and associativity need **coherences** for F 's unit and associativity: we need F to be a **wild (2,1)-functor**.

Slogan: for $F : \mathcal{C} \rightarrow \mathcal{C}$ **algebras** to be well-behaved up to level n , we need $\mathcal{C}$ and F to be well-behaved up to level **suc** n .

Impredicativity Gives the Weakly Terminal Cone

Build it in **levels**: dually, $\mu_n F$ for **algebras** and $\nu_n F$ for **coalgebras**, adding **coherence** each step.

$$\mu_0 F := \Pi_{X:\mathcal{C}} D X$$

$$\mu_1 F := \Sigma_{u:\mu_0 F} (u \text{ **natural**})$$

$$\mu_2 F := \Sigma_{u:\mu_1 F} (u \text{ **coherent**})$$

$$\vdots$$

$$\nu_0 F := \|\Sigma_{X:\mathcal{C}} D X\|_{\mathcal{U}}$$

$$\nu_1 F := \nu_0 F / (\text{**naturality**})$$

$$\nu_2 F := \nu_1 F // (\text{**coherence**})$$

$$\vdots$$

On the right, a **coequalizer** then a **coequifier** (higher inductive types, *not* 0-truncated **quotients**).

How can we find μF and νF from these? We can use $\mu_2 F$ and $\nu_2 F$ together with some categorical techniques: **splitting** idempotents.

Total (Co)cones & Idempotents

$\mu_2 F$ carries a **total cone**, dually $\nu_2 F$ a **total cocone**, of **coalgebra morphisms**:

$\text{iter} : \Pi_{(X,c)} ((\mu_2 F, \text{cons}) \rightarrow (X, c)), \quad \text{coiter} : \Pi_{(X,d)} ((X, d) \rightarrow (\nu_2 F, \text{des})).$

Their **apex morphisms** $e := \text{iter}(\mu_2 F, \text{cons})$ and $\text{coiter}(\nu_2 F, \text{des})$ are **idempotents** (by **naturality** of $(\text{co})\text{iter}$):

$$\begin{array}{ccc}
 & \mu_2 F & \\
 e \nearrow & \iota & \searrow e \\
 \mu_2 F & \xrightarrow{e} & \mu_2 F
 \end{array}$$

How coherent is $\iota : e \cdot e = e$?

- a **wild idempotent** is just the pair (e, ι) ;
- a **wild (2,1)-idempotent** also makes ι **coherent**: the two ways to collapse $e \cdot e \cdot e$ to e using ι agree (by **coherence** of $(\text{co})\text{iter}$).

Splitting Idempotents

Splitting is self-dual. We say $e : X \rightarrow X$ splits if we have a pair $r : X \rightarrow X'$ and $s : X' \rightarrow X$ with:

$$\begin{array}{ccc} & X' & \\ r \nearrow & & \searrow s \\ X & \xrightarrow{e} & X \end{array}$$

$$\begin{array}{ccc} & X & \\ s \nearrow & & \searrow r \\ X' & \xrightarrow{\text{id}_{X'}} & X' \end{array}$$

Lemma. If the apex morphism $e := \text{iter } \mu_2 F : \mu_2 F \rightarrow \mu_2 F$ splits, its retract is the initial algebra $(\mu F, \text{cons})$; dually a retract of the apex morphism of $\nu_2 F$ gives the terminal coalgebra $(\nu F, \text{des})$.

Idea: s turns the total (co)cone into one on X' that is the identity there, so the map out of X' is forced: weakly initial becomes initial.

Wild $(2, 1)$ -Idempotent Functions Split

So, does e split? In general a wild idempotent need not split.

Theorem (Shulman, after Lurie). Every wild $(2, 1)$ -idempotent $e : X \rightarrow X$ on a type $X : \mathcal{U}$ splits, via the sequential limit

$$X' := \Sigma_{x:\mathbb{N} \rightarrow X} \Pi_{n:\mathbb{N}} (e(x(\text{suc } n)) = x n).$$

Its extra coherence is what makes this infinite limit behave; the limit ranges over $\mathbb{N}$, which is why we need $\mathbb{N}$.

However, this result is about types, we need something for (co)algebras.

Lifting to (Co)algebras

Key Lemma. The forgetful functor $U_{\mathbf{Alg} F} : \mathbf{Alg} F \rightarrow \mathcal{U}$ lifts:

- splittings to splittings;
- weakly terminal wild $(2, 1)$ -cones of $U_{\mathbf{Alg} F}$ to wild cones of $\mathbf{Id}_{\mathbf{Alg} F}$.

Dual results hold for $U_{\mathbf{Coalg} F} : \mathbf{Coalg} F \rightarrow \mathcal{U}$.

So the apex wild $(2, 1)$ -idempotent on $\mu_2 F / \nu_2 F$ splits, and the retract inherits the (co)algebra structure, and is (co)initial.

Putting It Together

Impredicativity builds the (co)cones in levels:

- encoding $\rightsquigarrow \mu_0 F, \nu_0 F \rightsquigarrow$ weakly (co)initial: (co)iteration
- + naturality $\rightsquigarrow \mu_1 F, \nu_1 F \rightsquigarrow$ (co)initial with UIP
- + coherence $\rightsquigarrow \mu_2 F, \nu_2 F \rightsquigarrow$ a total wild $(2, 1)$ -(co)cone

The apex morphism at $\mu_2 F$ and $\nu_2 F$ is a wild $(2, 1)$ -idempotent; its underlying function splits in $\mathcal{U}$ (Shulman/Lurie), and the (co)cone and splitting lift to the (co)algebras.

So the retract is an initial algebra / terminal coalgebra.

Payoff: W-Types, M-Types, and More

The **polynomial** functor for $A : \text{Type}$ and $x : A \vdash B : \text{Type}$:

$$F_{A,x.B} X := \Sigma_{x:A} (B \rightarrow X)$$

is strict, hence a $(2, 1)$ -functor, so **impredicativity** supplies its **$(2, 1)$ -(co)cone** and we obtain:

- the **initial algebra**: the **inductive** type $W_{x:A} B$;
- the **terminal coalgebra**: the **coinductive** type $M_{x:A} B$.

The same recipe builds higher inductive types (in particular **coequalizers** and **coequifiers**, which we use when encoding **coinductive** types).

In type theory with **function extensionality**, and an **impredicative** universe.

Parametric Sizes

Sized Types: Termination & Productivity

Proof assistants only accept total functions: terminating recursion and productive corecursion both need to keep track of sizes.

Sized types track this in the types: annotate the inductive type μF / coinductive type νF with a size/ordinal i , giving approximations $\mu^i F$, $\nu^i F$ of the full (co)inductive types.

Termination and productivity then reduce to well-founded induction on the size.

Inductive Types: Initial Algebras

An inductive type is the initial algebra of a functor F ; approximate it by iterating F from 0:

$$0 \rightarrow F\,0 \rightarrow F^2\,0 \rightarrow \dots \rightarrow F^i\,0 \rightarrow \dots$$

Here $\mu^i F := F^i\,0$: terms constructed in i steps.

Examples.

$\mu X. 1 + X$	$\rightsquigarrow$	$\mathbb{N}$
$\mu X. 1 + A \times X$	$\rightsquigarrow$	lists of A
$\mu X. 1 + X + X^{\mathbb{N}}$	$\rightsquigarrow$	Brouwer ordinals

For Brouwer ordinals the chain only stabilises at ω_1 ; we need large sizes!

Coinductive Types: Final Coalgebras

Dually, a **coinductive** type is the **final coalgebra** of F ; approximate it by **iterating** F from 1:

$$\dots \rightarrow F^i 1 \rightarrow \dots \rightarrow F 1 \rightarrow 1$$

Here $\nu^i F := F^i 1$: behaviour distinguishable within i **destructor** steps.

Examples.

$\nu X. A \times X$	$\rightsquigarrow$	streams over A
$\nu X. 1 + A \times X$	$\rightsquigarrow$	possibly infinite lists over A
$\nu X. 1 + A \times X \times X$	$\rightsquigarrow$	possibly infinite binary trees over A

These chains stabilise sooner: typically at ω .

Agda's Largest Size is Inconsistent

How should the full μF , νF relate to their approximations $\mu^i F$, $\nu^i F$?

Agda adds a **largest size** ∞ (a *closure ordinal*) and sets

$$\mu F = \mu^\infty F, \quad \nu F = \nu^\infty F.$$

But then ∞ satisfies $\infty < \infty$ contradicting **irreflexivity** of a well-order.

This makes Agda's sized types **inconsistent**; we give a **consistent** alternative, which we show by building a model.

The Fix: Quantify over Sizes

Recover the full type with **quantifiers** instead of a top size:

$$\mu F := \exists_i \mu^i F, \quad \nu F := \forall_i \nu^i F.$$

Intuitively:

- an **inductive** term is **built** within **some** (**abstract**) **height**;
- a **coinductive** term is **observable** (**uniformly**) to any **depth**.

The catch: $\exists, \forall$ must be **parametric**, keeping the size **abstract**.

Using Σ, Π would make $\mu F, \nu F$ both **too large**.

In detail: The Type of Sizes

It looks like $\mathbb{N}$, but the **elimination** differs:

no pattern matching on sizes, so they never influence **computation**.

A primitive type **Size** with constructors

$$0_{\text{Size}} : \text{Size}, \quad \text{suc}_{\text{Size}} : \text{Size} \rightarrow \text{Size},$$

a preorder $i \leq j$ (a **proposition**), and $(i < j) := (\text{suc}_{\text{Size}} i \leq j)$.

The only **elimination** is **well-founded induction**, expressed using $\exists/\forall$.

Parametric Quantifiers

$\exists$ and $\forall$ over sizes are **impredicative**: although we have that **Size** is **large**,
 $\exists_i A : \mathcal{U}$ and $\forall_i A : \mathcal{U}$ are the size of A .

The **universal** behaves exactly like Π :

$$\forall_i A \simeq \Pi_{i:\text{Size}} A.$$

But the **existential** eliminates only into **small** types:

$$\exists_i A \not\approx \Sigma_{i:\text{Size}} A.$$

We can think of this as $\|\Sigma_{i:\text{Size}} A\|_{\mathcal{U}}$, the size stays **abstract**, and we cannot project it out.

We define bounded quantifiers:

$$\exists_{j < i} A := \exists_j ((j < i) \times A),$$

$$\forall_{j < i} A := \forall_j ((j < i) \rightarrow A).$$

Parametricity Axioms

For **small** $A, B[i], B[i, x], C[i]$, we **axiomatise** that these **canonical maps** showing how they compute over the other connectives are **equivalences**:

$$\exists_i \Pi_{x:A} \exists_{j < i} B[x, j] \rightarrow \Pi_{x:A} \exists_i B[x, i] \quad (\exists / \Pi)$$

$$\Sigma_{x:A} \forall_i B[x, i] \rightarrow \forall_i \Sigma_{x:A} B[x, i] \quad (\forall / \Sigma)$$

$$\exists_i \exists_{j < i} C[j] \rightarrow \exists_i C[i] \quad (\exists / <)$$

$$\forall_i C[i] \rightarrow \forall_i \forall_{j < i} C[j] \quad (\forall / <)$$

Note that the first equivalence only shows a **weak** commuting:
right-to-left it provides **suprema** of sizes, with *no* explicit limit operator.

This eventually lets us define the full **inductive** types.

Well-Founded Induction

The eliminator for **Size** is **well-founded induction**, a **fixpoint** operator:

$$\text{ind}^{\text{Size}} : \left(\forall_i \left((\forall_{j < i} A[j]) \rightarrow A[i] \right) \right) \rightarrow \forall_i A[i].$$

Its **β -rule** unfolds one step (kept **propositional**, for decidability):

$$\text{ind}^{\text{Size}} f\ i = f\ i\ (\lambda j < i. \text{ind}^{\text{Size}} f\ j).$$

Lemma (uniqueness of fixpoints). $\text{ind}^{\text{Size}} f$ is the **unique** term satisfying this unfolding: the type of such terms is **contractible**.

Building the Approximations

For $A : \text{Size} \rightarrow \mathcal{U}$ we will use $\diamond A := \lambda_i \exists_{j < i} A j$ and $\square A := \lambda_i \forall_{j < i} A j$.

In particular, we define $\mu^{(-)}F$, $\nu^{(-)}F$ as the **initial algebra** and **final coalgebra** of $F \circ \diamond, F \circ \square : (\text{Size} \rightarrow \mathcal{U}) \rightarrow (\text{Size} \rightarrow \mathcal{U})$, where

$$(F \circ \diamond) A := \lambda_i F (\exists_{j < i} A j),$$

$$(F \circ \square) A := \lambda_i F (\forall_{j < i} A j),$$

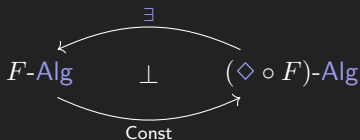
and where we construct $\mu^{(-)}F$, $\nu^{(-)}F$ using **well-founded induction**:

$$\mu^{(-)}F := \text{ind}^{\text{Size}} (\lambda_i \lambda_{X:\forall_{j < i} \mathcal{U}} (F \circ \diamond) X i) : \text{Size} \rightarrow \mathcal{U},$$

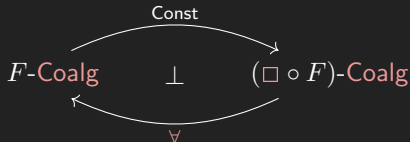
$$\nu^{(-)}F := \text{ind}^{\text{Size}} (\lambda_i \lambda_{X:\forall_{j < i} \mathcal{U}} (F \circ \square) X i) : \text{Size} \rightarrow \mathcal{U}.$$

Assembling the Full Type

Key Lemma. When F weakly commutes with $\exists$ — the canonical map $\exists_i F (\exists_{j < i} A j) \rightarrow F (\exists_i A i)$ is an equivalence — we have an adjunction:



When F weakly commutes with $\forall$ — the canonical map $F (\forall_i A i) \rightarrow \forall_i F (\forall_{j < i} A j)$ is an equivalence — we have an adjunction:



All (Co)inductive Types

Every **polynomial** functor F weakly commutes with $\exists$ and $\forall$.

Left adjoints preserve **initial** and right adjoints preserve **terminal**, so $\exists_i \mu^i F$ is **initial**, and $\forall_i \nu^i F$ is **terminal**.

Theorem. We have all **W-types** and all **M-types**.

Improvement: with **Size** = $\mathbb{N}$, we only reach **finitely-branching W-types**; our larger **Size** reaches **all** of them.

Consistency: A Realisability Model

We prove **consistency** with a **realisability** model (the assemblies of Hyland's effective topos). To interpret types with **computational content**, fix a model of computation (a **pca**) here Kleene's $\mathcal{K}_1$:

- **codes** are naturals $n : \mathbb{N}$, which can be seen as codes for the partial recursive functions $\varphi_n : \mathbb{N} \rightarrow \mathbb{N}$;
- **application** $m \cdot n := \varphi_m(n)$, a **partial** operation where we write $(m \cdot n) \downarrow$ when $m \cdot n$ **terminates**, and $(m \cdot n) \uparrow$ when $m \cdot n$ **diverges**.

Definition. An **assembly** $(X, \Vdash_X)$ is a set X with a relation $\Vdash_X \subseteq \mathbb{N} \times X$ such that every $x : X$ has at least one **realiser** $n \Vdash_X x$. A **morphism** $f : X \rightarrow Y$ is a function such that there **merely** exists a code e that tracks it:

$$n \Vdash_X x \text{ implies } (e \cdot n) \downarrow \text{ and } e \cdot n \Vdash_Y f x.$$

Maps are thus forced to be **computable (on codes)**.

The ∇ Inclusion Forces Parametricity

Any set X becomes an **assembly** $\nabla X := (X, \mathbb{N} \times X)$ by giving *every* code to *every* element.

This carries **no** computational data, so the elements are **indistinguishable** when defining **tracked** maps out of it.

This way we can model both **impredicativity** and **parametricity**:

- $\mathcal{U}$ will be interpreted as ∇ (the set of small types);
- **Size** will be interpreted as ∇ (a large enough set of ordinals).

Small Types: Modest Sets and PERs

Large types are **assemblies**; the **small** types need a **small** subcategory.

- a **modest assembly** is one where each code realises **at most one** element (the code *determines* the element);
- a **partial equivalence relation (PER)** is a symmetric and transitive $R \subseteq \mathbb{N} \times \mathbb{N}$, which gives a modest assembly $(\mathbb{N}/R, \in)$:

$$\text{PER} \simeq \text{Mod} \subset \text{Asm}.$$

The collection of all **PERs** is just a set, so we can form the **universe**

$$\mathcal{U} := \nabla \text{PER}.$$

Since ∇ makes the **PERs** **indistinguishable**, quantifying over *all* of them stays **small**: this is the model's **impredicativity**.

Sizes and Quantifiers in the Model

We interpret Size is a set of ordinals of larger cardinality than $|\mathcal{U}| = \omega$:

$$\text{Size} := \nabla \omega_1.$$

Earlier models based on relational parametricity with $\text{Size} := \mathbb{N}$ (Nuyts–Vezzosi–Devriese) only had **finitely branching inductive** types.

- $\forall_i A$ is the **tracked** dependent product $\prod_{i:\text{Size}} A$;
- $\exists_i A$ is the ***U*-truncation** of $\Sigma_{i:\text{Size}} A$: **reflecting** the **assembly** into a **PER** (left adjoint to $\text{Mod} \hookrightarrow \text{Asm}$), forgetting the **witness** so that the size stays **abstract**.

Since ∇ makes the **ordinals indistinguishable**, this satisfies the **parametricity** axioms.

Fixpoints

Well-founded induction ind^{Size} is interpreted by recursion on ω_1 ; it is tracked because every pca has a fixpoint combinator fix :

$$\text{fix } f \ a \simeq f \ (\text{fix } f) \ a,$$

which realises the unfolding β -rule.

So the model interprets the whole theory, and even validates impredicativity and UIP (although the syntax uses neither).

A cubical assemblies (assemblies internal to cubical sets) model should likewise validate univalence.

Conclusion

Two Constructions of (Co)inductive Types

To summarize:

	Impredicative universe	Parametric sizes
quantify over	all algebras	all sizes
small because	$\mathcal{U}$ is impredicative	$\exists, \forall$ are parametric
universal property	splitting idempotents	$\exists \dashv \text{Const}$ and $\text{Const} \dashv \forall$
key ingredient	$\mathbb{N}$	large sizes
compatible with	UIP / univalence	UIP / univalence

Two routes to one goal: initial algebras and terminal coalgebras for all polynomial functors.

Future Work

- higher **coinductive** types? I think the conclusion is that these either do not add anything new, or are inconsistent.
- make parametricity **compute**: bridges/cubical (Cavallo–Harper 2021), or **interval-free** internal parametricity (Altenkirch–Chamoun–Kaposi–Shulman 2024);
- a **classical** variant: an **infinite hierarchy** of universes with **Size** demoted to a **pre-type**. Are **parametric** $\exists/\forall$ consistent with **LEM**? Can we obtain **large elimination**?
- higher **inductive** types?

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